A Machine Learning Excursion to Cellular Automata

Exploring Machine Learning Models on Rule 90 and Game of Life

Introduction

Cellular Automata are fascinating toy systems where the emergence of complexity from simplicity can be explored. Rule 90 is a one-dimensional, binary cellular automaton where the state of each cell at the next time step is determined by the exclusive OR (XOR) of its two immediate neighbors' states. Game of Life is a popular CA on a $n \times n$ grid, where each cell is either alive or dead. Every step, cells with exactly two alive neighbors will maintain their state, cells with exactly three alive neighbors will become alive, and cells with any other number of neighbors will die.
We use RNN and LSTM to predict the very next generation of Rule 90, and CNNs on Life

Method

Rule 90

The following parameters were used for the network architecture in Rule 90 program.

Parameters	Values
sequence_length	20
num_samples	10000
embedding_dim	8
nn_units	32
batch_size	32

Game of Life

We used a simple CNN architecture. The convolution layer has only one filter of the size 3×3, we used adam optimizer and cross entropy loss function. We generated 10,000 samples, and varied grid size.

Conclusion

Learning Cellular Automata is very hard in general. We weren't able to get more than about 53% accuracy when using RNN or LSTM on Rule 90. CNN perform quite well on learning Life.

Future Plans

We can try comparing the performance of RNNs, LSTMs and CNNs when learning over a few generations for both the CA. We can extend our study on other CA such as Brian's Brain and maybe some physical systems like Ising Model also.

RNNs and LSTM networks perform poorly to predict next generation of Rule90. CNNs prove to be somewhat powerful for learning to predict the next generation of Game of Life. We also find that the results seem to get better with larger grid sizes.

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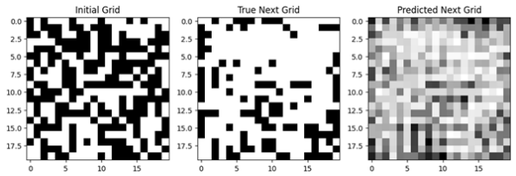

Figure 1: Predicted outcomes for Grid Size 20.

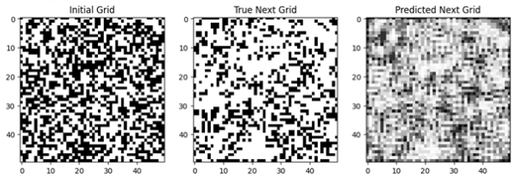

Figure 2: Predicted outcomes for Grid Size 50.

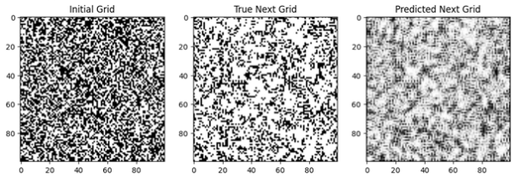

Figure 3: Predicted outcomes for Grid Size 100.

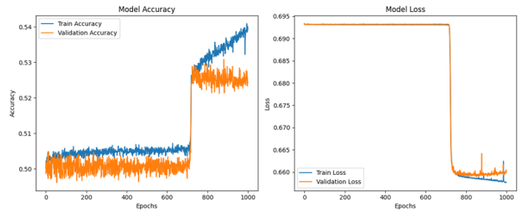

Figure 4: RNN, trained for 1000 epochs.

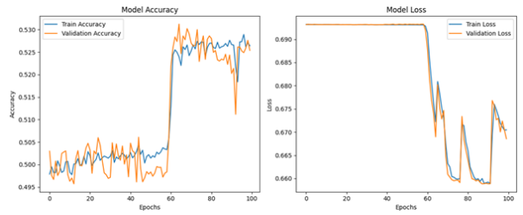

Figure 5: LSTM, trained for 100 epochs.